NUS National University of Singapore

Map, then state

What separates exact sum-product from projected gradient descent

BP–GD mechanism study · evidence checked through 6 November 2026

Two decoders

Held fixed

regular (10200, 5100) graph, degree 3 / 6

paired by frame ID: stored transmitted word, noise realization and channel LLR vector

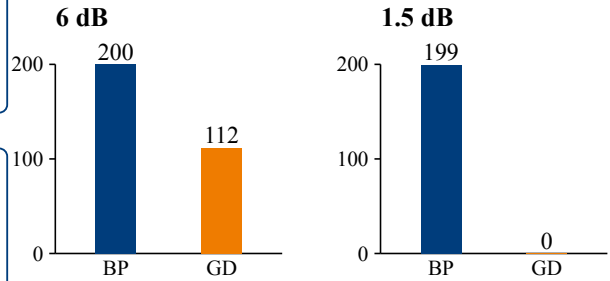
endpoint: transmitted word recovered

Exact sum-product (BP)

stop at first zero syndrome, cap 80

Projected gradient descent (GD)

30 updates on the compound objective



Paired by frame ID: BP recovers 88 more words at 6 dB and 199 more at 1.5 dB.

Evidence: RQ2-NATIVE-COMPOSITION-001 v2 ledger, rungs 1 and 11; 6 dB frames 60025–60224, 1.5 dB frames 60225–60424. Unless a different denominator is printed beside a value, counts are transmitted words recovered out of 200 fresh frames.

Is it what BP stores, or how BP updates?

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State

one scalar per bit,
or one scalar per directed variable-to-check edge

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Direction

downhill on an objective,
or the map applied

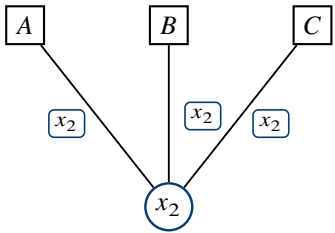
Is it what BP stores, or how BP updates?

State	Direction	Step
one scalar per bit, or one scalar per directed variable-to-check edge	downhill on an objective, or the map applied	filtered by a decrease test, or taken whole

A constructed bridge crosses state, direction and step. Its endpoint gates reproduce the retained transformed recurrence; the other cells are interventions, not native BP or native GD.

What is stored

Projected GD

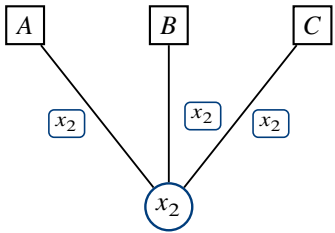


One scalar per bit persists. Every check reads the same x_2, so the bit cannot say different things to A, B and C.

Notation: L channel LLR, U check-to-variable message calculated inside F, Q variable-to-check message, $z = \tanh(Q/2)$ the factorial's stored edge coordinate, and x GD's scalar state in $[-1, 1]$.

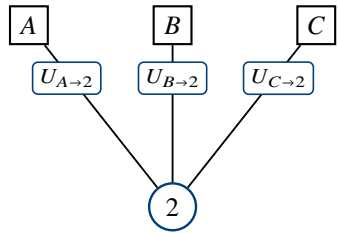
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Exact sum-product

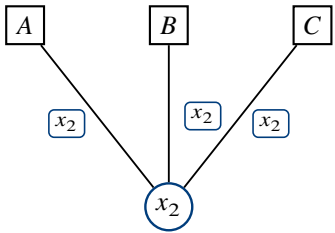


Native BP retains directed variable-to-check and check-to-variable messages. The factorial bridge represents its variable-to-check quantity by the stored coordinate $z_{2\to c} = \tanh(Q_{2\to c}/2)$.

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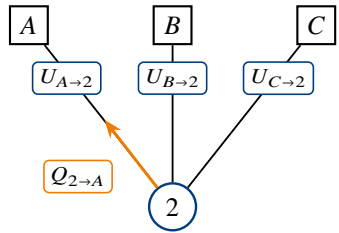
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Projected GD



One scalar per bit persists. Every check reads the same x_2, so the bit cannot say different things to A, B and C.

Exact sum-product



The persistent coordinate is $z_{2\to c} = \tanh(Q_{2\to c}/2)$. Inside F, $Q_{2\to A} = L_2 + U_{B\to 2} + U_{C\to 2}$ leaves out $U_{A\to 2}$, so each recipient can receive a different value.

Notation: L channel LLR, U check-to-variable message calculated inside F, Q variable-to-check message, $z = \tanh(Q/2)$ the factorial's stored edge coordinate, and x GD's scalar state in $[-1, 1]$.

How it is updated

Projected GD

$$\mathbf{x}^{+} = \Pi\left[\mathbf{x} - \alpha\, \mathbf{D}^{-1} \nabla J(\mathbf{x})\right]$$

evaluate J and calculate its partial derivatives

try $\alpha = 0.2 \cdot 2^{-b}$, $b = 0, 1, \ldots, 16$

project the trial into the box $[-1, 1]$

accept the first strict decrease of J

Π: box projection; $\mathbf{D}$: diagonal preconditioner.
No decrease: keep the state.

Held fixed: registered graph and each frame ID's stored channel LLR vector. GD uses the registered strict-decrease backtracking rule. F is the exact sum-product map in RQ2-COMMON-COORDINATE-STATE-UPDATE-001; the later H path count is from RQ2-BP-OBJECTIVE-PATH-001 on frames 70921–71120.

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Exact sum-product

$$\mathbf{z}^{+} = F(\mathbf{z})$$

evaluate the map F at the stored values

assign the result

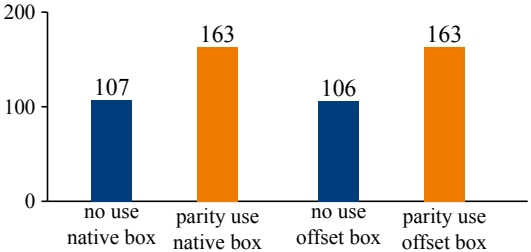
F is the sum-product map: the check operation, then the recipient-excluded variable calculation of the previous slide.

There is no step length and no H-decrease test. On the later sealed 1.5 dB path replay, the unit-map arm increased H on 1,327 of 6,000 updates.

Held fixed: registered graph and each frame ID's stored channel LLR vector. Uses the registered strict-decrease backtracking rule. F is the exact sum-product map in RQ2-COMMON-COORDINATE-STATE-UPDATE-001; the later H path count is from RQ2-BP-OBJECTIVE-PATH-001 on frames 70921–71120.

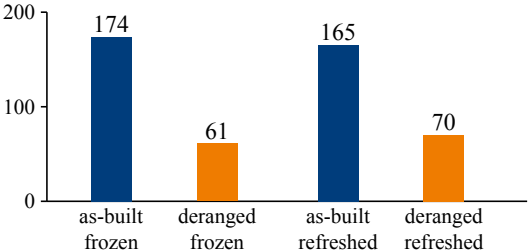
What the offset state actually changes

Offset use × projection interval



Parity use raises recovery from 107 to 163 with native bounds. Offset-induced bounds change 107 to 106 without parity use and 163 to 163 with it.

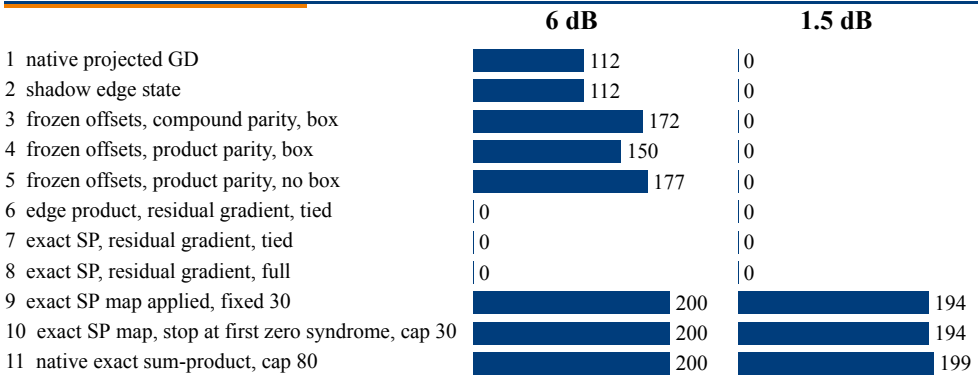
Check assignment × replacement



As-built assignment beats derangement under freezing and replacement. Replacement changes the as-built count from 174 to 165; the equivalence gate does not pass.

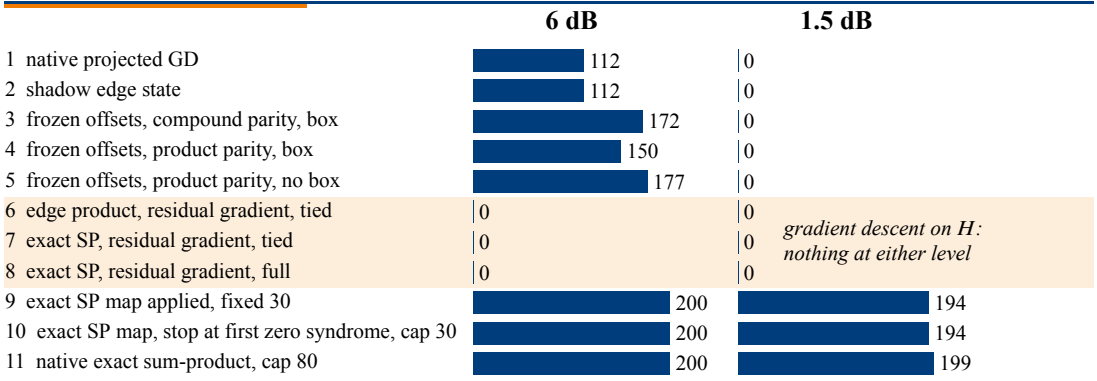
Complete-valid 6-dB evidence: RQ2-GD-OFFSET-SPLIT-001 (frames 30001–30200) and RQ2-GD-PARITY-ASSIGNMENT-REFRESH-001 (frames 40001–40200); counts are transmitted words recovered out of 200. These are constructed projected-gradient contrasts, not BP-message or low-SNR claims.

Historical ladder: one confounded jump



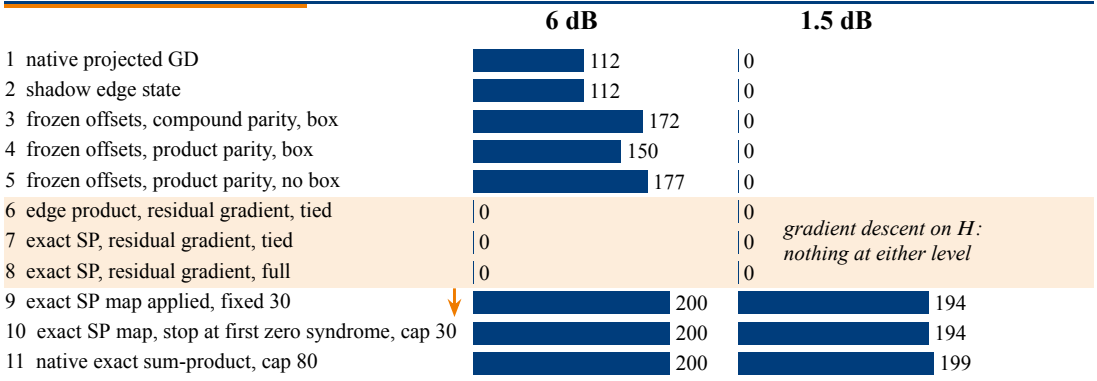
Evidence: RQ2-NATIVE-COMPOSITION-001 v2 rung registry; rungs 1–9 use 30 updates, rung 10 stops at the first zero syndrome with cap 30, and rung 11 is native exact sum-product with cap 80; 6 dB frames 60025–60224, 1.5 dB frames 60225–60424. Rungs 3–5 use frozen per-check offsets. The rung 5→6 transition changes state, target and update together, so it is not a one-factor causal contrast.

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Historical ladder: one confounded jump



Rung 8 to 9 changes the update: gradient step to direct map, recovering +200 at 6 dB and +194 at 1.5 dB.

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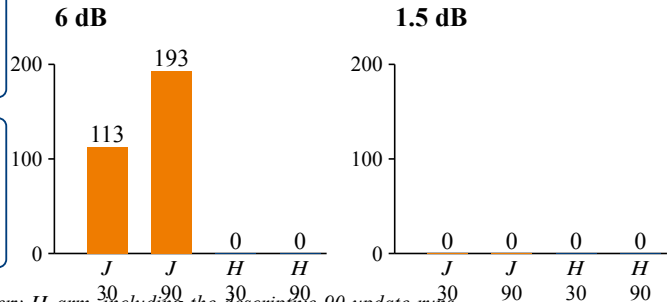
One objective

Held fixed

tied state: one scalar per bit
negative metric gradient
backtracking step, box
the bounded common coordinate

Changed: the objective only

J: the compound parity objective
$H(\mathbf{z}) = \frac{1}{2n}\,(\mathbf{z} - F(\mathbf{z}))^{\mathrm{T}}\mathbf{W}\,(\mathbf{z} - F(\mathbf{z}))$,
zero exactly at BP's fixed points

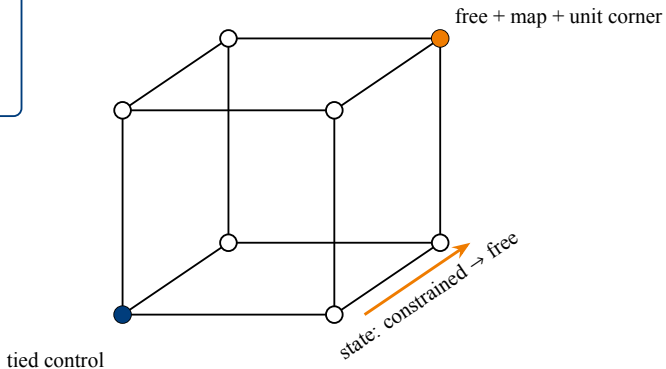


Changing J to H leaves recovery at zero in every H arm, including the descriptive 90-update runs.

Evidence: RQ2-COMMON-COORDINATE-TARGET-001 (30 updates) and the descriptive, non-inferential RQ2-COMMON-COORDINATE-BUDGET-90-001 diagnostic; 6 dB frames 61745–61944, 1.5 dB frames 61969–62168. $\mathbf{W}$ weights each edge by one over its bit's degree.

Eight arms

State
constrained: one scalar per bit, broadcast
free: one scalar per directed
variable-to-check edge persists



Held fixed in all eight cells: objective H, graph, frames, box, 30 updates, coordinate $z = \tanh(Q/2)$. Evidence: RQ2-COMMON-COORDINATE-STATE-UPDATE-001 v2, an outcome-independent registration; primary and replay passes equal bit for bit.

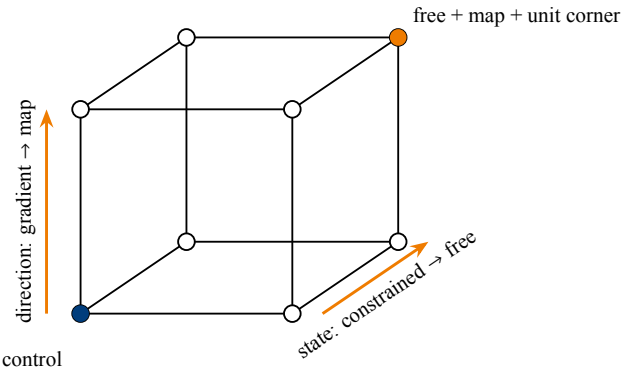
Eight arms

State

constrained: one scalar per bit, broadcast
free: one scalar per directed variable-to-check edge persists

Direction

gradient: negative metric gradient of H
map minus state: $F(\mathbf{z}) - \mathbf{z}$



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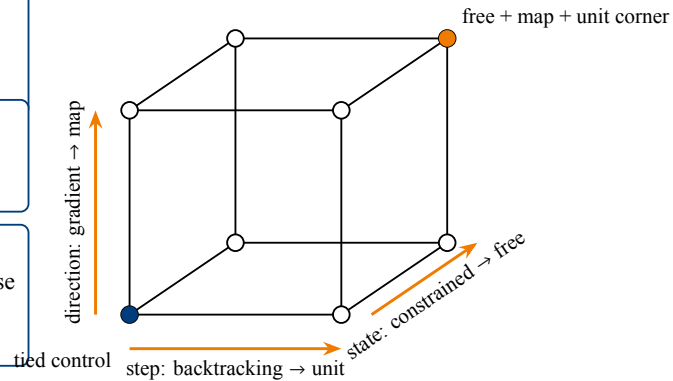
constrained: one scalar per bit, broadcast
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Direction

gradient: negative metric gradient of H
map minus state: $F(\mathbf{z}) - \mathbf{z}$

Step

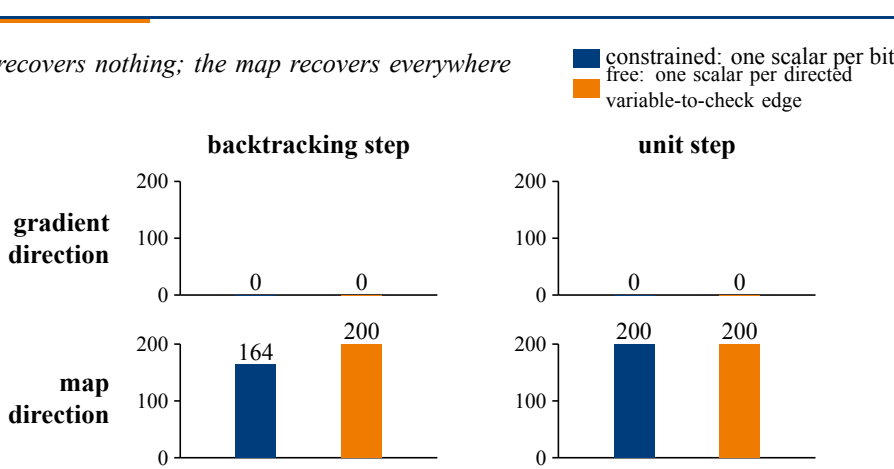
backtracking: $\alpha = 0.2 \cdot 2^{-b}$, strict decrease of H
unit: $\alpha = 1$, $F(\mathbf{z})$ assigned, no test



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Six decibels

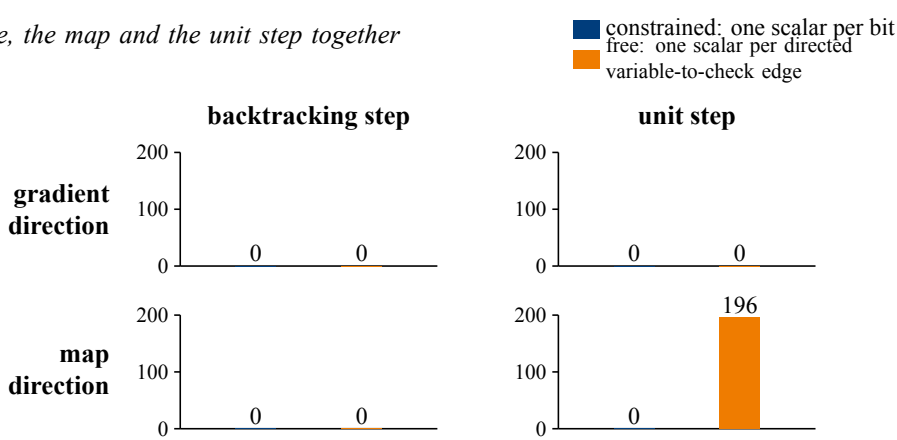
The gradient recovers nothing; the map recovers everywhere



Evidence: RQ2-COMMON-COORDINATE-STATE-UPDATE-001 v2, 6 dB transfer, frames 70721–70920; direction contrasts Holm-adjusted $p \le 7.7 \times 10^{-49}$.

One and a half decibels

Only free state, the map and the unit step together



Evidence: RQ2-COMMON-COORDINATE-STATE-UPDATE-001 v2, 1.5 dB transfer, frames 70921–71120; the nonzero cell versus each neighbour, Holm-adjusted $p = 2.4 \times 10^{-58}$.

Three contrasts

- The four true-gradient arms recover 0/200 at each SNR
- Map direction: recovers at 6 dB in every cell
- Free state under the gradient: nothing, at either level
- Free state under the map: +36 at 6 dB, +196 at 1.5 dB
- Unit step under the map: +36 at 6 dB, +196 at 1.5 dB
- Three-factor interaction on the risk-difference scale: −0.18 at 6 dB, +0.98 at 1.5 dB

Evidence: contrasts.csv and interactions.csv of RQ2-COMMON-COORDINATE-STATE-UPDATE-001 v2; paired risk differences on 200 frames per level, McNemar exact tests with Holm adjustment. The +36 at 6 dB is the backtracking cell, 164 to 200; the +196 at 1.5 dB is the unit cell, 0 to 196.

Zero is not a diagnosis

Observed at update 30

all four true-gradient arms recover 0/200 transmitted words at each SNR

$$G_{0.2}(q) = \frac{q - \Pi_{\mathscr{C}}\left(q - 0.2D^{-1}\nabla h(q)\right)}{0.2}$$

The endpoint diagnosis combines motion, its late trend, distance from $F(q) = q$, and what the constraints remove.

Candidate endpoint conditions

Budget-limited progress candidate
diagnostic and state change remain non-small and decrease

Wrong stationary point
small diagnostic and map displacement, wrong hard decision

Restricted
a constraint removes most of a non-small direction

Derivative terms cancel
combined derivative is small while its components are not

Stall, cycle, or ambiguous
motion stops decreasing, repeats, or lies in the gray band

The registered labels may overlap; mixed and ambiguous outcomes are retained

Registered definitions: RQ2-GD-TRAJECTORY-DIAGNOSIS-001; four retained true-gradient arms, 30-update endpoints, and prospective thresholds. These diagnostics route a follow-up; they do not themselves establish a decoding-performance cause.

At update 30: registered progress-candidate condition

Registered budget-limited-progress-candidate condition at update 30: the diagnostic and state change remain non-small, the late trend decreases, and no short cycle is detected.

True-gradient arm	6 dB	1.5 dB
one scalar per bit · backtracking	200 candidates	200 candidates
one scalar per bit · unit step	199 candidates; 1 stall/cycle	200 candidates
one scalar per directed variable-to-check edge · backtracking	200 candidates	200 candidates
one scalar per directed variable-to-check edge · unit step	200 candidates	200 candidates

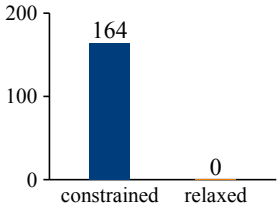
6 dB: 799/800 candidates; one stall/cycle. **1.5 dB:** 800/800 candidates.

This ruled against already-settled endpoints at update 30 and selected the objective-path test; it does not prove eventual recovery or that update budget caused the decoding gap.

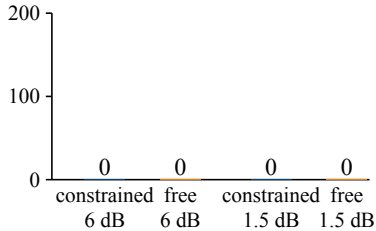
Evidence: RQ2-GD-TRAJECTORY-DIAGNOSIS-001 v2; 6 dB frames 70721–70920 and 1.5 dB frames 70921–71120; both transfers and the unpooled joint summary are complete-valid with exact primary/replay equality.

Edge copies alone

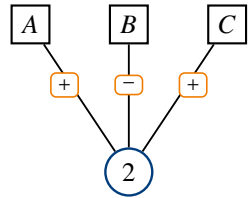
Compound objective, 6 dB



Residual objective



Relaxed copies

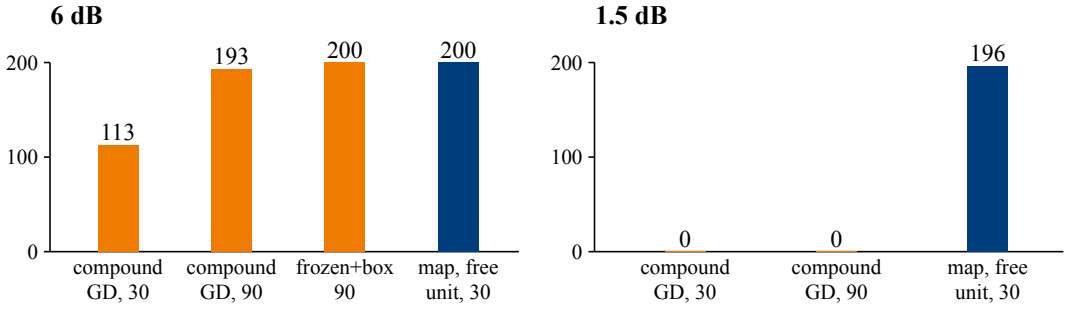


Across 200 frames, mean counts are 5098.24 of 5100 checks satisfied by the copies, 5031.30 by the hard-decision estimate, and 126.86 disagreeing edge signs.

In these registered gradient comparisons, releasing edge copies is harmful or null

Evidence: RQ2-BOUNDED-EDGE-FREEDOM-001, 6 dB frames 61521–61720, risk difference −0.82 [−0.87, −0.76]; RQ2-COMMON-COORDINATE-STATE-001, frames 70273–70472 and 70473–70672, risk difference 0.

Two levels



Evidence: 6 dB compound, RQ2-COMMON-COORDINATE-TARGET-001 and descriptive -BUDGET-90-001, frames 61745–61944; assigned/frozen arm, RQ2-CAUSAL-FACTORS-001 block A, frames 50025–50224. Related 1.5 dB native compound diagnostic, RQ2-COMMON-COORDINATE-BUDGET-90-001: 0/200 at updates 30 and 90. Map arm, RQ2-COMMON-COORDINATE-STATE-UPDATE-001, frames 70721–71120. Frame sets differ.

Same fixed points, different directions

Residual objective

$r(\mathbf{z}) = \mathbf{z} - F(\mathbf{z})$

$H(\mathbf{z}) = \frac{1}{2n} r(\mathbf{z})^{\mathrm{T}} \mathbf{W} r(\mathbf{z})$

Direct recurrence

$\mathbf{z}^{+} = F(\mathbf{z})$

direction $F(\mathbf{z}) - \mathbf{z} = -r(\mathbf{z})$

$$\nabla H(\mathbf{z}) = \frac{1}{n} (I - J_F(\mathbf{z}))^{\mathrm{T}} \mathbf{W} r(\mathbf{z})$$

The gradient direction contains a Jacobian-transpose transformation. The map direction does not. Both are zero at the same exact fixed points, but away from a fixed point they need not point in the same direction.

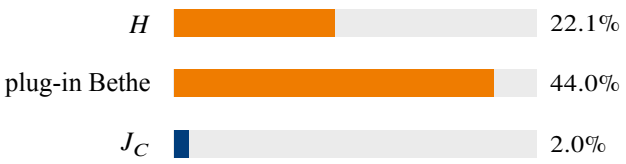
$r(\mathbf{z})$ → changes the direction → $-(I - J_F)^{\mathrm{T}} \mathbf{W} r / n$

Same fixed points do not imply the same trajectory, decrease property, or decoded word.
Definitions: RQ2-COMMON-COORDINATE-TARGET-001 and RQ2-COMMON-COORDINATE-STATE-UPDATE-001. F is the transformed exact sum-product recurrence, $\mathbf{W}$ is the registered fixed diagonal matrix, and J_F is the Jacobian of F. This algebra is a mechanism statement, not a decoder-performance result.

No registered objective supplied both tests

Successful exact-recurrence path at 1.5 dB

Fraction of 6,000 requested updates on which the candidate objective increases:



Counts: H 1,327; plug-in Bethe 2,640; J_C 120 increases, each out of 6,000 requested updates.

Endpoint-ascent caveat: all 196 terminal-success proxy frames reached their first centered-readout zero syndrome before any positive J_C endpoint change; 96/196 never increased over all 30 updates.

J_C passes the 5% path test

but its negative fixed-matrix gradient has median angle 83.4° from $F(\mathbf{z}) - \mathbf{z}$, with 2,156 cases where $F(\mathbf{z}) - \mathbf{z} = 0$ while the gradient-based candidate remains nonzero.

Decision

H and plug-in Bethe fail the path test. J_C fails the direction test.

The 2% count is therefore not an explanation for first validity. Because $J_C(\mathbf{z}) = J(C\mathbf{z})$ is flat on $\ker C$, a barrier must be certified in the quotient represented by the mean coordinates $\mathbf{u} = C\mathbf{z}$, with a separate no-jump check for finite projected trials.

Evidence: complete-valid RQ2-BP-OBJECTIVE-PATH-001 and RQ2-BP-OBJECTIVE-DIRECTION-001; derived post-outcome cross-tabulation on 1.5 dB frames 70921–71120. Exact-recurrence proxy, not native BP.

What the mathematics rules out

Ruled out exactly

$v_L(\mathbf{z}) = \mathbf{M}[\Phi_L(\mathbf{z}) - \mathbf{z}]$

has a nonsymmetric Jacobian witness for every one of the 400 registered LLR vectors.

Therefore no exact C^2 potential generates this field under the registered bounded edge coordinates and fixed diagonal matrix.

Still geometrically possible

A symmetric positive-definite matrix exists pointwise for all active states:

464/464 at 6 dB,
3,564/3,564 at 1.5 dB.

The condition-bound counts are 461/464 and 3,561/3,564.

First constructive family: complete-valid scientific failure. The registered one-hop graph-local fit selected identity scaling only. Held-frame median angles remained 83.48° at 6 dB and 82.95° at 1.5 dB.

Pointwise existence is not a decoder or BER result. The failed one-hop family is closed.

Evidence: RQ2-BP-MAP-INTEGRABILITY-001 v2 (400/400 certificates), RQ2-ACTIVE-METRIC-FEASIBILITY-001 v1, and RQ2-GRAPH-LOCAL-METRIC-001 v1. Scope: current coordinates, registered fixed matrix, and the exact fitted one-hop family only.

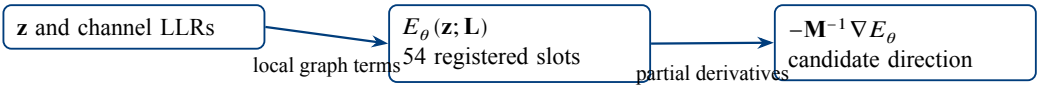
Factorized objective: development fit

Held fixed

one bounded scalar per directed variable-to-check edge; fixed diagonal matrix; initialization; box; centered incident-edge mean readout; 30 updates

Changed

one objective function may use check-local and recipient-excluded two-hop dependencies. The optimization variables are not BP messages.



25 coefficients are fitted; 29 are fixed exactly to zero. Exact sum-product is an offline teacher during development and is forbidden during candidate inference.

6 dB development median angle 83.46° → 39.74°; candidate p90 41.72°; relative error 0.658.

1.5 dB development median angle 82.99° → 65.75°; candidate p90 72.94°; relative error 0.927.

HISTORICAL DEVELOPMENT RESULT: 400 reused frames; no decoder or performance claim.

Evidence: RQ2-GRAPH-FACTORIZED-ENERGY-001 Stage A v2; frames 70721–70920 and 70921–71120, 4,028 active states. The sealed artifact contains 54 slots: 25 fitted and 29 exact zero. Primary and normalized replay science SHA-256 are equal.

Fresh direction screen

48 fresh direction cases; coefficients unchanged; primary and replay identical. This screen calculated directions only. It did not run a decoder.

6 dB: PASS 24 frames, 53 states

median angle $39.59^\circ \leq 60^\circ$
p90 angle $41.38^\circ \leq 85^\circ$
median relative error $0.659 \leq 0.9$

1.5 dB: FAIL 24 frames, 453 states

median angle $66.16^\circ > 60^\circ$
p90 angle $74.01^\circ \leq 85^\circ$
median relative error $0.938 > 0.9$

The direction generalized at 6 dB but missed two prospective low-SNR gates. The exact graph-factorized objective-function family is closed; its decoder stages did not run.

This is a causal routing result about direction alignment, not a decoding-performance result.

Evidence: RQ2-GRAPH-FACTORIZED-ENERGY-001 Stage B v1; fresh frames 71569–71592 and 71593–71616. Operational status complete-valid; scientific status direction_screen_fail.

Which candidate is best?

Best verified 6 dB gradient result

compound objective + box
90 updates: 200/200 transmitted words
30 updates: 177/200

Verified at 6 dB only

Best current cross-SNR candidate to test

state-dependent two-hop symmetric positive-definite matrix
fitted on retained 6 dB and 1.5 dB trajectories
no fresh direction screen or decoder result

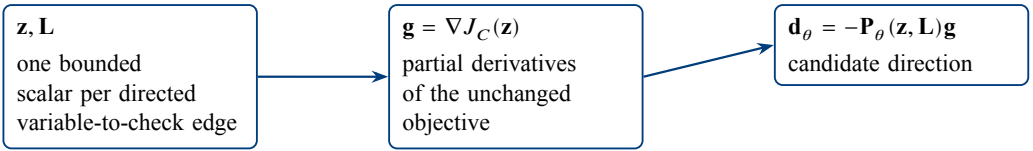
Unvalidated as a decoder

On that 200-case 6 dB population, the first result proves that a registered gradient construction can close the recovery gap with a larger update budget. It does not solve the low-SNR problem: that exact construction has no completed 1.5 dB transfer, while a related native compound diagnostic remained 0/200 through 90 updates.

Therefore "best verified result" and "best candidate for the two-SNR research objective" are different statements

Evidence: RQ2-CAUSAL-FACTORS-001 block A, 6 dB frames 50025–50224, assigned/frozen compound+box at checkpoints 30 and 90; related 1.5 dB result from RQ2-COMMON-COORDINATE-BUDGET-90-001. The candidate status comes from RQ2-TWO-HOP-POSITIVE-METRIC-001 V5.

Best current gradient candidate to test



5,636 synthetic assertions = 2,240 earlier source-and-interface assertions + 3,396 dormant Stage-C assertions. A pure checker recalculates every objective, partial-derivative vector, direction, trial, Armijo decision, readout, syndrome, and cross-record link; 35 altered traces and constructor-bypassed coefficients fail closed.

Production-size synthetic receipt: four candidate/control cells, three one-thread repetitions each. Update-cap trace 69,861,148 bytes; decoder 5.78–6.45 s; complete check 5.53–5.76 s; measured child peak RSS at most 899,579,904 bytes.
Boundary: source/resource readiness only—no complete-valid second range-audit result, registered frame, Stage-B science, retained-wall result, capsule audit, registered decoder-anchor execution, recovery, or BER/FER result. RQ2-TWO-HOP-POSITIVE-METRIC-001; support terms are not a new decoder name.

Projection can erase raw descent

Raw matrix direction

$\mathbf{d} = -\mathbf{P}_\theta \mathbf{g}$

$\mathbf{g}^T\mathbf{d} < 0$

(-2.8535×10^{-4})

Componentwise box

$\delta_b = \Pi_{[-1,1]}(\mathbf{z}+2^{-b}\mathbf{d})-\mathbf{z}$

All 13 trials reject

$\mathbf{g}^T\delta_b > 0$

$b = 0, \ldots, 12$

Nonzero two-hop candidate stall

registered fixed-diagonal projected-gradient mapping:
$\|\mathbf{G}_{0.2}\|_2 = 0.0180151$
diagnostic step: $\Delta J_C = -4.6152 \times 10^{-5}$
not box-stationary

Exact-zero control stall

same graph and channel-LLR vector
projected-gradient mapping is bit-exact zero
box-stationary under this diagnostic

Positive definiteness proves descent before projection; ordinary coordinatewise projection need not preserve it for a non-diagonal matrix action.

An independent dense implementation reproduces the 30-update and late-stall paths within 3×10^{-13}; a nondegenerate fixture proves that the recipient-excluded operation creates edge-specific state that affects the next direction.

Evidence: dormant six-variable synthetic fixture only. No registered frame, BP value, truth word, decoder rule, coefficient, threshold, or gate changed. This is a projection/action compatibility diagnosis, not performance evidence.

Full synthetic chain fails closed

synthetic graph, bounded state, channel LLRs, coefficients → unchanged J_C, fixed diagonal, candidate direction → offline comparison diagnostic and origin adapter → three-row reduction and registered gates

Exact interface identities: the frozen comparison vector equals the candidate direction, and all three synthetic origin rows repeat one already-materialized diagnostic.

Target-specific gate: pass

the intended source, pairing and denominator contracts hold

Generic direction + combined gates: fail

$0.6852959854888271^\circ < 15^\circ$ required paired improvement

The earlier supplied-arithmetic seam produced 0.9935439809818444°; it is distinct from this full chain, which calculates the unchanged J_C path before the offline comparison. The 15° requirement was not changed.

This is deterministic interface validation only—not BP agreement, population science, a decoder, or a performance result.

Evidence: 84 full-chain assertions within the pre-Stage-C subtotal of 2,240. This chain opens no registered frame and performs no state update, projection, line search, readout, recovery or BER/FER scoring, or threshold change.

V5 retained fit selects two-hop geometry

Complete-valid retained fit: 400 reused frames; exact primary/replay identity; production-constructor handoff passed; fresh cases opened: 0.

Registered retained fit loss



$[\theta_0, \beta_0, \beta_1, \beta_2, \beta_3] =$
$[0, -0.0989697, -0.191296, -0.705845, -0.00368998]$ all four two-hop coefficients are nonzero

This is a retained direction-fit result. It is not a fresh angle, transmitted-word recovery, or BER result.

Evidence: RQ2-TWO-HOP-POSITIVE-METRIC-001 Stage A V5; retained frames 70721–70920 and 70921–71120; terminal classification retained_fit_complete_nonzero_two_hop; science SHA-256 b14d0afb…c3f52c5.

Fresh scientific validation has not opened

V5 retained Stage A **complete-valid** → coefficient intake **verified** → v10 audit: **failed**, v11 retry: **failed** → V2 seal + amendment **complete**; retained wall + capsule audit **not complete**

Governance boundary: V1 remains invalid and excluded. The separately approved V2 seal is complete-valid (authorization `06944b79…803b`; manifest `a71ebef9…38c5`; member `173a68f4…3078`) and opened no frame.

Reserved fresh direction screen

6 dB: registered 24-case block
1.5 dB: registered 24-case block
24 cases at each level

Current permissions

Stage B: false
fresh input: false
registered decoder-anchor execution: false
BER/FER scoring: false

The approved corrected v11 retry completed the repository scan and failed on two registered gates: one documentary self-collision and a coordinator field of 6,140,592 KiB versus the 2,097,152-KiB cap. Fresh-child reproduction proved that field inherited a pre-existing process-lifetime peak across `execve`; it is not scanner allocation. No scientific input opened.

Formal failure stands; V11 result SHA-256 10618db4…a9f5ff. Fresh route parser: 54,668 KiB; threshold CSV: 80,024 KiB; controlled pre-exec peak: 269,952 KiB. All other registered gates passed; no manifest, fresh input, or Stage-B science exists.

Causal validation ladder

1. Direction
24+24 untouched cases
all gates at both levels

→

2. Decoder anchor
24+24 cases
candidate $\geq 20/24$

→

3. Recovery proxy
200+200 cases
candidate $\geq$ $190/200$

→

4. Native BP
same case identities
recovery and coded-bit BER

Causal contrast: the candidate and its zero-coefficient lower endpoint differ only in the registered coefficients: the candidate uses $\mathbf{P}_\theta(\mathbf{z}, \mathbf{L})$, while the lower endpoint uses $\mathbf{P}_0(\mathbf{z}, \mathbf{L})$. Every other registered operation stays fixed.

Recovery proxy: the decoder-anchor and transfer gates compare the candidate with the exact recurrence under the registered 30-update construction. Passing them establishes only the preregistered two-SNR recovery proxy.

A BP-approach claim still requires direct native sum-product comparison under its first-valid-syndrome stopping rule and 80-message-passing-iteration cap, reporting transmitted-word recovery and coded-bit BER.
Prospective gates: RQ2-TWO-HOP-POSITIVE-METRIC-001 experiment card, Stages B–D. Any claim remains bounded to the registered regular $(10200, 5100)$ code, AWGN populations, two SNRs and stated work limits.

Where the explanation stands

Map first; then state and step. In the 30-update factorial, every true-gradient arm recovered $0/200$ at both SNRs. At 1.5 dB, only free recipient-specific state, map displacement and a unit step together recovered transmitted words: $196/200$.

Why the gradient route differs. Its Jacobian-transformed direction is not the map displacement. Fixed-matrix candidates and the first graph-factorized objective did not supply the required low-SNR direction.

Best current candidate. Dormant source applies the current-state two-hop symmetric positive-definite matrix to unchanged-J_C partial derivatives, then uses the registered projection and Armijo rule. A synthetic late stall is not box-stationary because projection turns raw descent into ascent; the method remains unchanged and fresh direction and decoder-anchor gates remain unopened.

Current conclusion. A gradient-based decoder approaching native BP remains mathematically possible and experimentally unproven. No recovery or BER result approaching BP has yet been established. **The research objective is materially advanced, not complete.**

The next scientific evidence is the untouched direction screen—after every governance and source gate is valid.

This conclusion preserves interactions and does not allocate the native BP–GD gap additively. It distinguishes a verified 6 dB gradient result from an unvalidated cross-SNR candidate.